UNIX.

2026-03-28

Linux, , .
, ( ), .

. ~/abc1 april may:

```
eakara@fedora:~$ touch abc1
 cp abc1 april
 cp abc1 may
```

. april may monthly:

```
eakara@fedora:~$ mkdir monthly
 cp april may monthly
```

. monthly/may june:

```
eakara@fedora:~$ cp monthly/may monthly/june
 ls monthly
april  june  may
```

. monthly monthly.00:

```
eakara@fedora:~$ mkdir monthly.00
 cp -r monthly monthly.00
```

. monthly.00 /tmp

```
eakara@fedora:~$ cp -r monthly.00 /tmp
```

. april july :

```
eakara@fedora:~$ cd
 mv april july
```

. **july** **monthly.00:**

```
eakara@fedora:~$ mv july monthly.00
 ls monthly.00
july  monthly
```

. **monthly.00** **monthly.01**

```
eakara@fedora:~$ mv monthly.00 monthly.01
```

. **monthly.01** **reports:**

```
eakara@fedora:~$ mkdir reports
 mv monthly.01 reports
```

, . **reports/monthly.01** **reports/monthly:**

```
eakara@fedora:~$ mv reports/monthly.01 reports/monthly
```

, chmod, ls -l, ,

```
eakara@fedora:~$ touch file1
eakara@fedora:~$ chmod 644 file1
eakara@fedora:~$ ls -l file1
-rw-r--r--. 1 eakara eakara 0 мар 26 01:08 file1
```

, chmod, ls -l, ,

```
eakara@fedora:~$ touch file2
eakara@fedora:~$ chmod 700 file2
eakara@fedora:~$ ls -l file2
-rwx------. 1 eakara eakara 0 мар 26 01:11 file2
```

, chmod, ls -l, ,

```
eakara@fedora:~$ mkdir dir1
eakara@fedora:~$ chmod 751 dir1
eakara@fedora:~$ ls -ld dir1
drwxr-x--x. 1 eakara eakara 0 мар 26 01:14 dir1
```

~/may :

```
eakara@fedora:~$ cd
 touch may
 ls -l may
 chmod u+x may
 ls -l may
-rw-r--r--. 1 eakara eakara 0 мар 26 01:19 may
-rwxr--r--. 1 eakara eakara 0 мар 26 01:19 may
```

~/may :

```
eakara@fedora:~$ chmod u-x may
 ls -l may
-rw-r--r--. 1 eakara eakara 0 мар 26 01:19 may
```

monthly :

```
eakara@fedora:~$  chmod g-r,o-r monthly
```

~/abc1 **:**

```
eakara@fedora:~$ cd
 touch abc1
 chmod g+w abc1
```

:

```
eakara@fedora:~$ mount
/dev/sda3 on / type btrfs (rw,relatime,seclabel,compress=zstd:1,ssd,space_cache=v2,subvolid=256,subvol=/root)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=2315244k,nr_inodes=578811,mode=755,inode64)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64,usrquota)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=600,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsdelegate,memory_recursiveprot)
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=935708k,nr_inodes=819200,mode=755,inode64)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=42,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=6880)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,seclabel,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabel)
tmpfs on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,seclabel,size=1024k,nr_inodes=1024,mode=700,inode64,noswap)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,seclabel,nr_inodes=1048576,inode64,usrquota)
/dev/sda3 on /home type btrfs (rw,relatime,seclabel,compress=zstd:1,ssd,space_cache=v2,subvolid=257,subvol=/home)
/dev/sda2 on /boot type ext4 (rw,relatime,seclabel)
tmpfs on /run/credentials/systemd-resolved.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,seclabel,size=1024k,nr_inodes=1024,mode=700,inode64,noswap)
VMshare on /media/sf_VMshare type vboxsf (rw,nodev,relatime)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw,relatime)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=467852k,nr_inodes=116963,mode=700,uid=1000,gid=1000,inode64)
gvfsd-fuse on /run/user/1000/gvfs type fuse.gvfsd-fuse (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
portal on /run/user/1000/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
```

:

```
eakara@fedora:~$ cat /etc/fstab

#
# /etc/fstab
# Created by anaconda on Mon Mar  2 21:23:58 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=44e78210-be19-41a3-aaf3-c7bfa0d30fa1 / btrfs subvol=root,compress=zstd:1 0 0
UUID=5993d248-0e50-4644-a213-bd2f126e21e5 /boot ext4 defaults 1 2
UUID=44e78210-be19-41a3-aaf3-c7bfa0d30fa1 /home btrfs subvol=home,compress=zstd:1 0 0
```

:

```
eakara@fedora:~$ df
Файловая система 1К-блоков Использовано Доступно Использовано% Смонтировано в
/dev/sda3         81785856     13624992 65216608           18% /
devtmpfs           2315244            0  2315244            0% /dev
tmpfs              2339268           92  2339176            1% /dev/shm
tmpfs               935708         1452   934256            1% /run
tmpfs                 1024            0     1024            0% /run/credentials/systemd-journald.service
tmpfs              2339268            8  2339260            1% /tmp
/dev/sda3         81785856     13624992 65216608           18% /home
/dev/sda2          1992552       381956  1489356           21% /boot
tmpfs                 1024            0     1024            0% /run/credentials/systemd-resolved.service
VMshare          189141096    173891928 15249168           92% /media/sf_VMshare
tmpfs               467852          152   467700            1% /run/user/1000
```

fsck ( ) . :

```
eakara@fedora:~$ fsck /dev/sda1
fsck from util-linux 2.41.1
e2fsck 1.47.3 (8-Jul-2025)
fsck.ext2: Отказано в доступе while trying to open /dev/sda1
You must have r/w access to the filesystem or be root
eakara@fedora:~$
```

,

:

/usr/include/sys/io.h equipment.

```
eakara@fedora:~$ cp /usr/include/sys/io.h ~/equipment
```

~/ski.plases.

```
eakara@fedora:~$ mkdir ~/ski.plases
```

equipment ~/ski.plases.

```
eakara@fedora:~$ mv ~/equipment ~/ski.plases/
```

~/ski.plases/equipment ~/ski.plases/equiplist.

```
eakara@fedora:~$ mv ~/ski.plases/equipment ~/ski.plases/equiplist
```

abc1 ~/ski.plases, equiplist2.

```
eakara@fedora:~$ touch ~/abc1
eakara@fedora:~$ cp ~/abc1 ~/ski.plases/equiplist2
```

equipment ~/ski.plases.

```
eakara@fedora:~$ mkdir ~/ski.plases/equipment
```

~/ski.plases/equiplist equiplist2 ~/ski.plases/equipment.

```
eakara@fedora:~$ mv ~/ski.plases/equiplist ~/ski.plases/equiplist2 ~/ski.plases/equipment
```

~/newdir ~/ski.plases plans.

```
eakara@fedora:~$ mkdir ~/newdir
eakara@fedora:~$ mv ~/newdir ~/ski.plases/plans
```

---

chmod, ,
, , :

chmod

```
eakara@fedora:~$ mkdir australia
eakara@fedora:~$ chmod 744 australia
```

chmod

```
eakara@fedora:~$ mkdir play
eakara@fedora:~$ chmod 711 play
```

chmod

```
eakara@fedora:~$ touch my_os
eakara@fedora:~$ chmod 544 my_os
```

chmod

```
eakara@fedora:~$ touch feathers
eakara@fedora:~$ chmod 644 feathers
```

,

:

/etc/password.

```
eakara@fedora:~$ cat /etc/passwd
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
openvpn:x:990:990:OpenVPN:/etc/openvpn:/usr/sbin/nologin
nm-openvpn:x:989:989:Default user for running openvpn spawned by NetworkManager:/:/usr/sbin/nologin
abrt:x:173:173::/etc/abrt:/usr/sbin/nologin
wsdd:x:987:987:Web Services Dynamic Discovery host daemon:/:/usr/sbin/nologin
nm-openconnect:x:986:986:NetworkManager user for OpenConnect:/:/usr/sbin/nologin
rtkit:x:985:985:RealtimeKit:/:/usr/sbin/nologin
pipewire:x:984:984:PipeWire System Daemon:/run/pipewire:/usr/sbin/nologin
flatpak:x:982:982:Flatpak system helper:/:/usr/sbin/nologin
geoclue:x:981:981:User for geoclue:/var/lib/geoclue:/sbin/nologin
```

~/feathers ~/file.old.

```
eakara@fedora:~$ cp ~/feathers ~/file.old
```

~/file.old ~/play.

```
eakara@fedora:~$ mv ~/file.old ~/play/
```

~/play ~/fun.

```
eakara@fedora:~$ cp -r ~/play ~/fun
```

~/fun ~/play games

```
eakara@fedora:~$ mv ~/fun ~/play/games
```

~/feathers .

```
eakara@fedora:~$ chmod u-r ~/feathers
```

~/feathers cat. ,

```
eakara@fedora:~$ cat ~/feathers
cat: /home/eakara/feathers: Отказано в доступе
```

~/feathers. ,

```
eakara@fedora:~$ cp ~/feathers ~/copy_test
cp: невозможно открыть '/home/eakara/feathers' для чтения: Отказано в доступе
```

~/feathers .

```
eakara@fedora:~$ chmod u+r ~/feathers
```

~/play .

```
eakara@fedora:~$ chmod u-x ~/play
```

~/play. , .

```
eakara@fedora:~$ cd ~/play
bash: cd: /home/eakara/play: Отказано в доступе
```

~/play .

```
eakara@fedora:~$ chmod u+x ~/play
eakara@fedora:~$ cd ~/play
eakara@fedora:~/play$ cd
```

---

man mount, fsck, mkfs, kill , .

mount . mount (, ,) Linux. :

```
eakara@fedora:~$ sudo mount /dev/sdb1 /mnt/usb
```

fsck - . fsck . :

```
eakara@fedora:~$ sudo fsck /dev/sdb1
```

mkfs - . . :

```
eakara@fedora:~$ sudo mkfs.ext4 /dev/sdb1
```

```
eakara@fedora:~$ sudo mkfs.vfat /dev/sdb1
```

kill - . . . :

```
eakara@fedora:~$ kill -15 1234
```

```
eakara@fedora:~$ kill -9 1234
```

Linux, , .
, ( ), .